

Western Blot Integrity Atlas

Full reference-proteome molecular-weight audit report with embedded figures and complete tables

Defensive use only. This report helps evaluate whether a claimed western blot result is plausible and what controls are needed. It must not be used to plan, enable, or disguise blot substitution or fabrication.

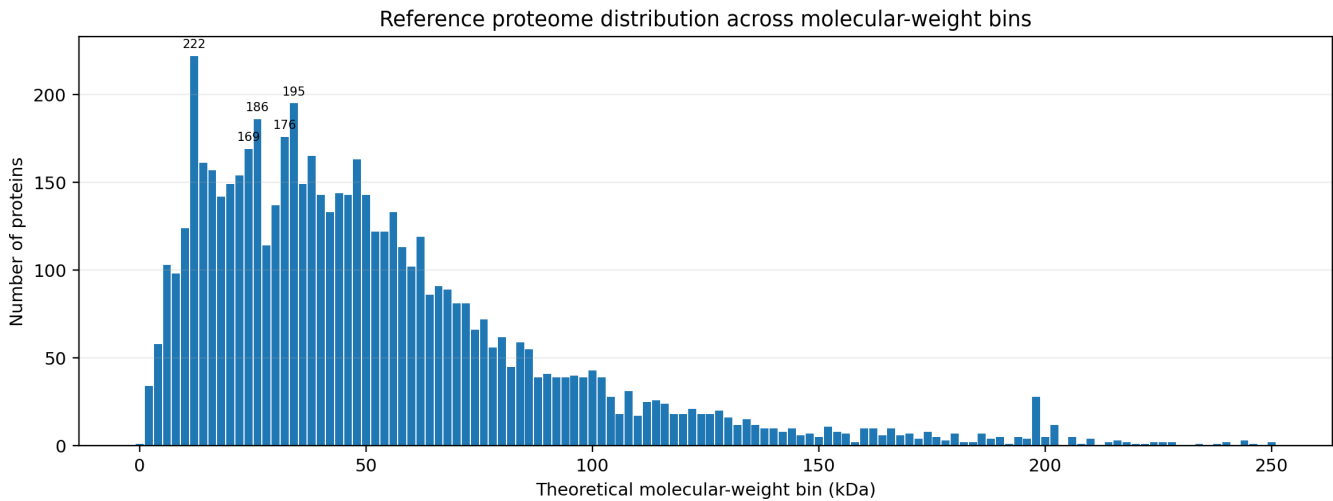
Metric	Value
Proteins parsed	6,066
Molecular-weight bins	140
Bin width	2 kDa
MW range	1.98 to 559.31 kDa
Median MW	44.87 kDa
Mean MW	54.78 kDa
Proteins <= 50 kDa	3,420 (56.4%)
Proteins 50-100 kDa	1,934 (31.9%)
Proteins > 100 kDa	712 (11.7%)
High-priority bins	14
Elevated-priority bins	21
Routine-priority bins	105
Source FASTA	UP000002311_559292.fasta

Method in one breath

For each protein sequence, the script computes theoretical average molecular weight from residue masses, assigns a molecular-weight bin, applies the external marker, signaling-prefix, and family-pattern annotations, maps supplied public malpractice cases to bins, and calculates an audit-burden priority. Similar apparent molecular weight is never treated as proof of identity; it only increases the need for full-blot provenance and orthogonal controls.

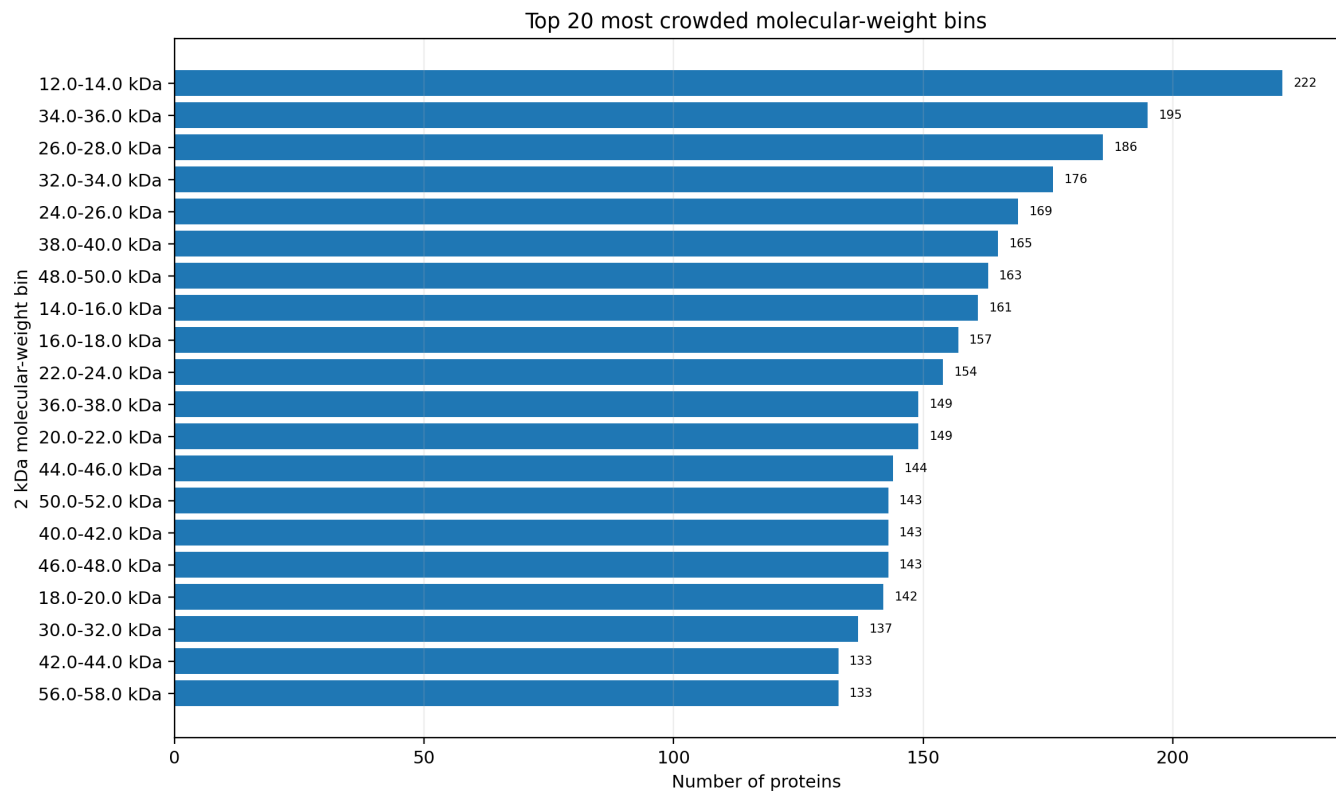
Molecular-weight distribution of the reference proteome.

Proteins were grouped into theoretical molecular-weight bins. Dense gel regions are especially relevant for western blot review because many unrelated proteins occupy nearby molecular-weight neighborhoods.



Most crowded molecular-weight neighborhoods.

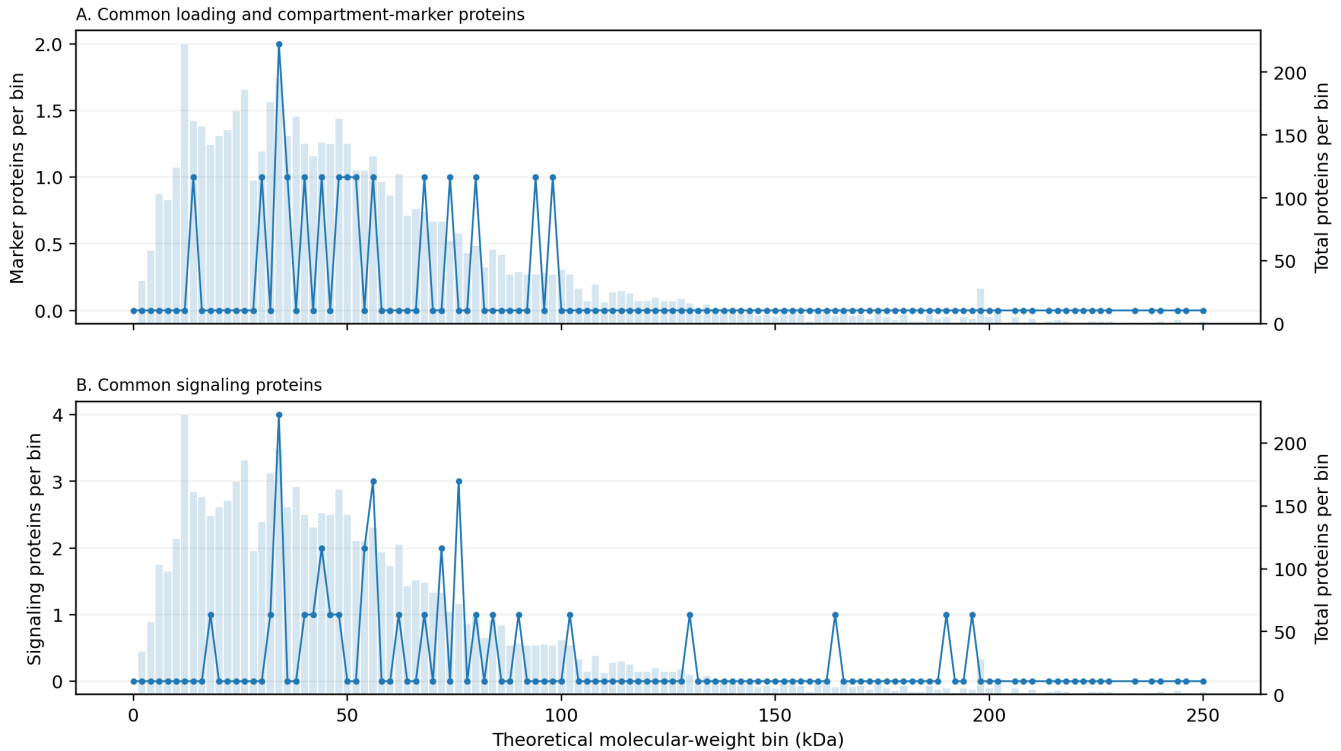
The top bins are ranked by protein count. These bins are not suspicious by themselves; they indicate where molecular weight alone is a weak identity criterion and where validation documentation becomes more important.



Distribution of common western blot markers and signaling proteins across molecular-weight bins.

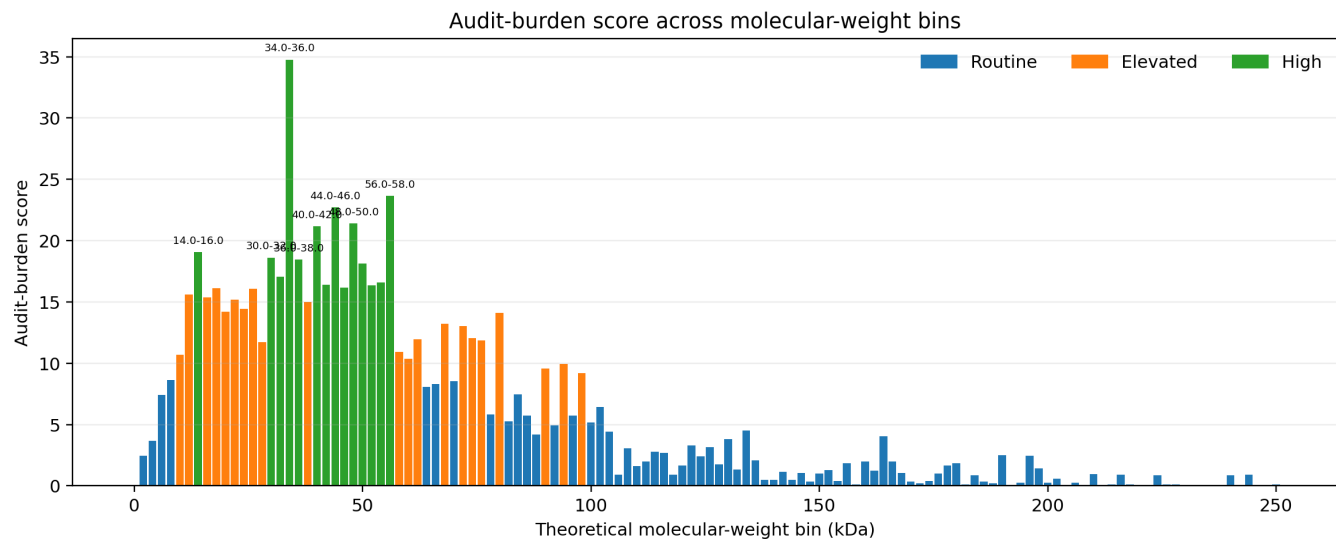
The pale background bars show total protein density, while the overlaid lines show audit-relevant annotations. Panel A shows configured loading or compartment markers; Panel B shows configured signaling-prefix matches.

Audit-relevant annotations across molecular-weight bins



Audit-burden score across molecular-weight bins.

The score integrates protein density, marker count, signaling-prefix count, and configured family/context information. High-scoring bins define high-documentation zones for western blot identity claims.



Highest-audit molecular-weight neighborhoods: score summary

MW bin	N	Markers	Signal	Priority	Score
34.0-36.0 kDa	195	2	4	high	34.8
56.0-58.0 kDa	133	1	3	high	23.6
44.0-46.0 kDa	144	1	2	high	22.7
48.0-50.0 kDa	163	1	1	high	21.4
40.0-42.0 kDa	143	1	1	high	21.1
14.0-16.0 kDa	161	1	0	high	19.1
30.0-32.0 kDa	137	1	0	high	18.6
36.0-38.0 kDa	149	1	0	high	18.4
50.0-52.0 kDa	143	1	0	high	18.1
32.0-34.0 kDa	176	0	1	high	17.1
54.0-56.0 kDa	122	0	2	high	16.6
42.0-44.0 kDa	133	0	1	high	16.4
52.0-54.0 kDa	122	1	0	high	16.4
46.0-48.0 kDa	143	0	1	high	16.1
18.0-20.0 kDa	142	0	1	elevated	16.1
26.0-28.0 kDa	186	0	0	elevated	16.1
12.0-14.0 kDa	222	0	0	elevated	15.6
16.0-18.0 kDa	157	0	0	elevated	15.3
22.0-24.0 kDa	154	0	0	elevated	15.2
38.0-40.0 kDa	165	0	0	elevated	15.0
24.0-26.0 kDa	169	0	0	elevated	14.4
20.0-22.0 kDa	149	0	0	elevated	14.2
80.0-82.0 kDa	62	1	1	elevated	14.1
68.0-70.0 kDa	89	1	1	elevated	13.2
72.0-74.0 kDa	81	0	2	elevated	13.1

Highest-audit molecular-weight neighborhoods

These bins combine protein density with configured markers, signaling-prefix matches, and family-rich regions. They are not substitution recommendations; they are places where reviewers should be most insistent about source files and controls.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
34.0-36.0 kDa	195	2	4	Actin_tubulin; Autophagy; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Mitochondrial_membrane; Proteasome; Protein_kinases; Protein_phosphatases; RAS_PKA_SNF1; TOR_AGC_signaling; Ubiquitin_system	high	ODC2, GCN3, TY3A-G, LST8 [TOR_AGC_signaling], COX11 [Mitochondrial_membrane], OAZ1, GID10, CYT1, CDC28 [Cell_cycle; Protein_kinases], YOR385W, CDC34 [Cell_cycle; Ubiquitin_system], SPC34, YNL194C, PHO4, SPE4, TY3A-I, AIM46, ECM11
56.0-58.0 kDa	133	1	3	Autophagy; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Histones; MAPK_signaling; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; Ubiquitin_system	high	YHL012W, CYS4, ERG4, RUD3, DIT2, AMF1, RIO1 [Protein_kinases], TPS1, FDC1, RCM1, FMP42, ARO8, MMT1, MDM10, ARH1, CLB2 [Cell_cycle], STB3, EHD3
44.0-46.0 kDa	144	1	2	Actin_tubulin; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Histones; MAPK_signaling; Proteasome; Protein_kinases; Protein_phosphatases; RAS_PKA_SNF1	high	RRP43, TYS1, DOM34, ARP2 [Actin_tubulin], DCC1, SEN2, CNS1 [Heat_shock_chaperones], MVD1, PEX27, CLB6 [Cell_cycle], MRP4, OLA1, PTC4 [Protein_phosphatases], TAZ1, VMA5, TAM41, TPK2 [Protein_kinases; RAS_PKA_SNF1], SKI8
48.0-50.0 kDa	163	1	1	Actin_tubulin; Autophagy; Cell_cycle; Histones; MAPK_signaling; Mitochondrial_membrane; Proteasome; Protein_kinases; Ubiquitin_system	high	TY4A-J, SRO9, DSE3, ZNG1, DUS1, RMA1, COT1, PEA2, VPS4, PMI40, RAM1, SIM1, IDP1, SPR28, TY4A-H, MEI4, RPT5 [Proteasome], LIF1
40.0-42.0 kDa	143	1	1	Actin_tubulin; Autophagy; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Histones; MAPK_signaling; Protein_kinases; Protein_phosphatases; Ubiquitin_system	high	ELO2, IBD2, RAV2, PDB1, RNR4, YML119W, LYS12, SAE2, YCR102C, DON1, RCL1, SCW4, DUG3, SRT1, TES1, URE2, MDG1, YBR053C
14.0-16.0 kDa	161	1	0	Autophagy; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Ribosomal_proteins; Ubiquitin_system	high	YER189W, YEL008W, YHL018W, YMR075C-A, COX16 [Mitochondrial_membrane], CDC26 [Cell_cycle], YMR082C, YLR302C, BOL2, ATP14 [Mitochondrial_membrane], Q0144, YMR122C, ABM1, YGL088W, RPL26A [Ribosomal_proteins], AIM4, RPL26B [Ribosomal_proteins], HTB2 [Histones]
30.0-32.0 kDa	137	1	0	Actin_tubulin; Autophagy; Cell_cycle; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Proteasome; Protein_phosphatases; Ubiquitin_system	high	ATO3, MRPL24, DCI1, CYC3, MGM101, BMH1, SPO7, ERS1, PRO3, MAM33, DAL80, LUC7, YCL002C, ATG27 [Autophagy], PIR5, YCL068C, RSM26, SDH2
36.0-38.0 kDa	149	1	0	Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Mitochondrial_membrane; Proteasome; Protein_kinases; Protein_phosphatases; RAS_PKA_SNF1	high	YMR262W, PHO92, YRB2, GPM2, YKL091C, OTU2, YKR075C, YPL039W, RRP36, RPC34, RFC4, IES2, LIA1, MRH1, YOR292C, RPA43, ELO1, MTD1
50.0-52.0 kDa	143	1	0	Actin_tubulin; Autophagy; Cell_cycle; Heat_shock_chaperones; Histones; Proteasome; Protein_kinases; Protein_phosphatases	high	PZF1, UBX7, TEF1, ARP6 [Actin_tubulin], TRM11, RDS2, BNA3, RTT109 [Histones], DAL1, TEP1, RAX1, RNY1, RBA50, VBA3, CLP1, COR1, MOD5, CUS1
32.0-34.0 kDa	176	0	1	Actin_tubulin; Autophagy; Cell_cycle; ER_UPR; Glycolytic_controls; Heat_shock_chaperones; Mitochondrial_membrane; Ribosomal_proteins; Ubiquitin_system	high	YLR271W, TDA4, YFT2, PRM4, SNZ3, CRD1, VPS71, SNZ2, PEX18, PCL7 [Cell_cycle], LNP1 [ER_UPR], YMR087W, ASK1, PIN2, YNG2, INM2, ATP10 [Mitochondrial_membrane], PCL1 [Cell_cycle]
54.0-56.0 kDa	122	0	2	Actin_tubulin; Autophagy; Cell_cycle; Heat_shock_chaperones; Histones; MAPK_signaling; Mitochondrial_membrane; Proteasome; Protein_kinases; Protein_phosphatases; Ubiquitin_system	high	LPD1, RNT1, OST1, GPA1, MST1, FAR10, TRM13, ITT1 [Ubiquitin_system], YMR265C, YMR027W, URN1, RTG3, NAR1, IRC15, RSC6, APE4, TIM54 [Mitochondrial_membrane], UGA2
42.0-44.0 kDa	133	0	1	Actin_tubulin; Cell_cycle; Heat_shock_chaperones; Histones; MAPK_signaling; Mitochondrial_membrane; Proteasome; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	high	PDE1, TOM40 [Mitochondrial_membrane], BMT6, MIG2, ETR1, CPR6, ARC1, RIM2, YOR338W, MTG1, SPG5, CAK1 [Protein_kinases], SAM2, SPC42, A15_BETA, PEX7, BCD1, TAF4
52.0-54.0 kDa	122	1	0	Actin_tubulin; Cell_cycle; Glycolytic_controls; Histones; Mitochondrial_membrane; Protein_kinases; Ubiquitin_system	high	SPS22, CLD1, YHR177W, CDC40 [Cell_cycle], YKL075C, YSP3, KTR6, VHS1 [Protein_kinases], HRB1, COX10 [Mitochondrial_membrane], SAD1, YLH47, SHM2, TMS1, YMR209C, RAD57, YPL245W, MCH2
46.0-48.0 kDa	143	0	1	Autophagy; Cell_cycle; ER_UPR; Glycolytic_controls; Histones; Proteasome; Protein_kinases; RAS_PKA_SNF1; TOR_AGC_signaling; Ubiquitin_system	high	KTR1, PSG1, AAT2, OPI1, TPK1 [Protein_kinases; RAS_PKA_SNF1], SPN1, CAR2, DMA1 [Ubiquitin_system], BDH2, NSE4, RNR2, YNL165W, RTC4, BAG7, INN1, SPE2, DDR48, ERV46
18.0-20.0 kDa	142	0	1	Actin_tubulin; Autophagy; Cell_cycle; ER_UPR; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	elevated	BUD25, YAR068W, YPL073C, IGO1, IRC23, YGL041W-A, YFL051C, GNA1, YHR180W, APS1, YPI1, FYV7, YLR281C, POP6, CAF20, RPL21A [Ribosomal_proteins], HED1, RPL21B [Ribosomal_proteins]
26.0-28.0 kDa	186	0	0	Cell_cycle; ER_UPR; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Proteasome; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	elevated	OCA6 [Protein_phosphatases], SNO4, CAT5, MGE1, TAE1, CYT2, MND1, YSA1, SHU2, YPT52, LSB1, SPR2, YKL035W-A, YOR283W, RIB5, TPT1, RTR1, VPS24
12.0-14.0 kDa	222	0	0	Actin_tubulin; Autophagy; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Ribosomal_proteins	elevated	YGL193C, RPS25B [Ribosomal_proteins], RTC3, YDR115W, RPS25A [Ribosomal_proteins], YIL054W, YLR012C, YMR141C, YBR209W, YOR072W, RPL33A [Ribosomal_proteins], FPR1, RPL33B [Ribosomal_proteins], YDL012C, YOR343C, YLR296W, CYC1, YPR014C
16.0-18.0 kDa	157	0	0	Actin_tubulin; Autophagy; Cell_cycle; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Proteasome; Protein_kinases; Ribosomal_proteins; Ubiquitin_system	elevated	RPS15 [Ribosomal_proteins], GAT3, YER145C-A, MRP49, SLM6, YGL108C, RPS23A [Ribosomal_proteins], RPS23B [Ribosomal_proteins], YCR025C, SRB7, EDC2, YAE1, MOH1, SWS2, GPI19, MTC7, MRPL51, DSS4
22.0-24.0 kDa	154	0	0	Autophagy; Cell_cycle; ER_UPR; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Proteasome; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	elevated	FRQ1, COA1 [Mitochondrial_membrane], SRL1, FUN14, CIN4, SDH3, HAM1, ADD37, BET3, YOR387C, ERV2, YLR179C, VEL1, YDR248C, YMR295C, ATG31 [Autophagy], YPL257W, RPL16A [Ribosomal_proteins]
38.0-40.0 kDa	165	0	0	Actin_tubulin; Glycolytic_controls; Heat_shock_chaperones; Histones; Proteasome; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	elevated	SHE1, YML037C, LSP1, HUT1, ARI1, SOR2, TFB3, DFM1, SOR1, GRE2, YER156C, ARV1, SUA7, RFC3, YMR315W, ARA2, PHA2, GTT3
24.0-26.0 kDa	169	0	0	Autophagy; Cell_cycle; ER_UPR; Glycolytic_controls; Heat_shock_chaperones; Proteasome; Ribosomal_proteins; Ubiquitin_system	elevated	FMP32, FAR3, FAU1, ERP2, SLX9, SBA1 [Heat_shock_chaperones], ERV25 [ER_UPR], DOT5, RCN1, THO1, TRP1, MRX11, YEL076C-A, YLR464W, UBC1 [Ubiquitin_system], YBR197C, PSF1, RCR2
20.0-22.0 kDa	149	0	0	Actin_tubulin; Autophagy; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Mitochondrial_membrane; Proteasome; Ribosomal_proteins; Ubiquitin_system	elevated	YGR273C, FYV6, JJJ3, IRC25 [Heat_shock_chaperones; Proteasome], YGL242C, MRPL36, VPS68, UPS1, YOL024W, SCM4, YGL262W, YNL208W, MPO1, FMP33, YKR045C, BNA1, MEH1, SDH4
80.0-82.0 kDa	62	1	1	Cell_cycle; Heat_shock_chaperones; Histones; MAPK_signaling; Protein_kinases; Ubiquitin_system	elevated	YGL114W, AVT4, MDV1, GFA1, HDA1 [Histones], FRE2, GSY2, RIM4, KRE6, YMR1, EXO1, UTP8, YOR365C, FRE5, YBR259W, APL1, NDC80, TRS85
68.0-70.0 kDa	89	1	1	Heat_shock_chaperones; Protein_kinases; Protein_phosphatases	elevated	ECM3, PTR2, DBP9, RCK2 [Protein_kinases], TPO3, MNT4, LYP1, MAL12, DIP5, MAL32, PEX8, MPH2, RIM101, MAL31, YKR078W, CTM1, XKS1, RL1
72.0-74.0 kDa	81	0	2	Actin_tubulin; Autophagy; Cell_cycle; Heat_shock_chaperones; MAPK_signaling; Protein_kinases; RAS_PKA_SNF1; Ubiquitin_system	elevated	UTP5, MDM32, CUE3, SNF1 [Protein_kinases; RAS_PKA_SNF1], ORC3, SPC72 [Actin_tubulin], SCC4, ELM1 [Protein_kinases], A15_ALPHA, FUI1, SPO1, RSM22, ALE1, MUB1, GRC3, RSC4, NEL1, MUS81
74.0-76.0 kDa	66	1	0	Cell_cycle; ER_UPR; Heat_shock_chaperones; Protein_kinases; Protein_phosphatases	elevated	SPH1, RPC82, ARE2, YNL115C, TPA1, NDC1, PIG1 [Protein_phosphatases], TRF5, MTO1, YJR098C, SGV1 [Protein_kinases], KAR9, CDC45 [Cell_cycle], YML082W, VPS52, TOP3, MET4, MAE1
62.0-64.0 kDa	119	0	1	Cell_cycle; ER_UPR; Histones; Protein_kinases; Protein_phosphatases; Ubiquitin_system	elevated	CLN1 [Cell_cycle], HNM1, PCM1, GLE1, YCK2, TOS3 [Protein_kinases], DAK2, PUS1, MET7, DAK1, DED81, LCB1, PRI2, YDL180W, UTP7, GIP2, GGA1, TDP1
76.0-78.0 kDa	72	0	3	Heat_shock_chaperones; Protein_kinases; Protein_phosphatases; TOR_AGC_signaling; Ubiquitin_system	elevated	YML020W, CBP1, YPL109C, HOF1, YBL111C, MSS116, PTR3, CRZ1, MAG2 [Ubiquitin_system], TMN2, LAM5, BPL1, ALK2 [Protein_kinases], HEH2, RRI2, PSO2, PAN3, YPK1 [Protein_kinases; TOR_AGC_signaling]
28.0-30.0 kDa	114	0	0	Actin_tubulin; Autophagy; Heat_shock_chaperones; Mitochondrial_membrane; Proteasome; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	elevated	SCL1 [Proteasome], ISY1, RPS0A [Ribosomal_proteins], YKL071W, IML3, USE1, RPL8B [Ribosomal_proteins], RPL8A [Ribosomal_proteins], LST7, MRPL15, RRP15, IES3, PSY3, SHE2, TAD2, RKI1, GRX3, PUP1 [Proteasome]
58.0-60.0 kDa	113	0	0	Autophagy; Cell_cycle; ER_UPR; Heat_shock_chaperones; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases	elevated	MGR3, CDC6 [Cell_cycle], ALG2, FCY21, THi6, GCR2, STP1, SDS23, HOM3, YJR015W [ER_UPR], GAL3, ANP1, FCY2, PLM2, DRN1, YPS6, PD11 [ER_UPR], WSC3

Common loading/compartiment markers present in this proteome file

MW kDa	Gene	Accession	Entry	Protein	MW bin
15.36	HHT1	P61830	H3_YEAST	Histone H3	14.0-16.0 kDa
30.43	POR1	P04840	VDAC1_YEAST	Non-selective voltage-gated ion channel 1	30.0-32.0 kDa
34.47	NOP1	P15646	FBRL_YEAST	rRNA 2'-O-methyltransferase fibrillarin	34.0-36.0 kDa
35.75	TDH3	P00359	G3P3_YEAST	Glyceraldehyde-3-phosphate dehydrogenase 3	34.0-36.0 kDa
36.85	ADH1	P00330	ADH1_YEAST	Alcohol dehydrogenase 1	36.0-38.0 kDa
41.69	ACT1	P60010	ACT_YEAST	Actin	40.0-42.0 kDa
44.74	PGK1	P00560	PGK_YEAST	Phosphoglycerate kinase	44.0-46.0 kDa
49.80	TUB1	P09733	TBA1_YEAST	Tubulin alpha-1 chain	48.0-50.0 kDa
50.92	TUB2	P02557	TBB_YEAST	Tubulin beta chain	50.0-52.0 kDa
53.94	HXK2	P04807	HXKB_YEAST	Hexokinase-2	52.0-54.0 kDa
57.52	ZWF1	P11412	G6PD_YEAST	Glucose-6-phosphate 1-dehydrogenase	56.0-58.0 kDa
69.66	SSA1	P10591	HSP71_YEAST	Heat shock protein SSA1	68.0-70.0 kDa
74.47	KAR2	P16474	BIP_YEAST	Endoplasmic reticulum chaperone BIP	74.0-76.0 kDa
81.41	HSP82	P02829	HSP82_YEAST	ATP-dependent molecular chaperone HSP82	80.0-82.0 kDa
95.53	VPB1	P32563	VPP1_YEAST	V-type proton ATPase subunit a, vacuolar isoform	94.0-96.0 kDa
99.62	PMA1	P05030	PMA1_YEAST	Plasma membrane ATPase 1	98.0-100.0 kDa

Complete molecular-weight bin summary, low to high

All molecular-weight bins are shown. Long Families and Examples / audit families cells wrap and continue across pages.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
0.0-2.0 kDa	1	0	0		routine	YJR151W-A
2.0-4.0 kDa	34	0	0	Ribosomal_proteins	routine	YNL048C-A, YOR302W, YPR160W-A, YHR073C-B, YJL077W-A, YMR182W-A, YMR242W-A, YJL136W-A, YGL007A, YHL015W-A, YAR035C-A, RPL41A [Ribosomal_proteins], RPL41B [Ribosomal_proteins], YKL145W-A, YNR075C-A, YFR012W-A, YGR174W-A, YBL039C-A
4.0-6.0 kDa	58	0	0	Mitochondrial_membrane	routine	YJL077W-B, YMR315W-A, YBR221W-A, YMR272W-B, YGL006W-A, MFA2, YML007C-A, YDR118W-A, YER053C-A, PMP1, YBR296C-A, YLR264C-A, YGR204C-A, YJL052C-A, PLS1, YBL100W-C, YKL106C-A, YLR364C-A
6.0-8.0 kDa	103	0	0	ER_UPR; Mitochondrial_membrane; Ribosomal_proteins	routine	MC06, YLR154C-G, YGR146C-A, YML054C-A, YOR293C-A, YDR194W-A, YOR072W-B, PMP3, YBR072C-A, YLR157W-E, YOR376W-A, YBR121C-A, YML100W-A, YBR196C-A, YER078W-A, Q0017, YCR095W-A, YIL046W-A
8.0-10.0 kDa	98	0	0	Autophagy; Heat_shock_chaperones; Mitochondrial_membrane; Ribosomal_proteins; Ubiquitin_system	routine	YGR121W-A, ER11, COX17 [Heat_shock_chaperones; Mitochondrial_membrane], HSK3, EGO2, DAD4, ATG44 [Autophagy], YLR157W-D, KSH1, CMC4, ATX1 [Heat_shock_chaperones], TFB5, YNL162W-A, HUB1 [Ubiquitin_system], RPB10, YKL183C-A, YGR035W-A, VMA9
10.0-12.0 kDa	124	0	0	Heat_shock_chaperones; Histones; Mitochondrial_membrane; Proteasome; Ribosomal_proteins; Ubiquitin_system	elevated	YGR161W-C, YSF3, LSM3, YFR009W-A, ACB1, STF1, YJL197C-A, RPL43A [Ribosomal_proteins], RPL43B [Ribosomal_proteins], YPL119C-A, TOS6, TIM9 [Mitochondrial_membrane], WIP1, RSM19, TIM10 [Mitochondrial_membrane], SME1, YLR379W-A, SEM1 [Proteasome]
12.0-14.0 kDa	222	0	0	Actin_tubulin; Autophagy; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Ribosomal_proteins	elevated	YGL193C, RPS25B [Ribosomal_proteins], RTC3, YDR115W, RPS25A [Ribosomal_proteins], YIL054W, YLR012C, YMR141C, YBR209W, YOR072W, RPL33A [Ribosomal_proteins], FPR1, RPL33B [Ribosomal_proteins], YDL012C, YOR343C, YLR296W, CYC1, YPR014C
14.0-16.0 kDa	161	1	0	Autophagy; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Ribosomal_proteins; Ubiquitin_system	high	YER189W, YEL008W, YHL018W, YMR075C-A, COX16 [Mitochondrial_membrane], CDC26 [Cell_cycle], YMR082C, YLR302C, BOL2, ATP14 [Mitochondrial_membrane], Q0144, YMR122C, ABM1, YGL088W, RPL26A [Ribosomal_proteins], AIM4, RPL26B [Ribosomal_proteins], HTB2 [Histones]
16.0-18.0 kDa	157	0	0	Actin_tubulin; Autophagy; Cell_cycle; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Proteasome; Protein_kinases; Ribosomal_proteins; Ubiquitin_system	elevated	RPS15 [Ribosomal_proteins], GAT3, YER145C-A, MRP49, SLM6, YGL108C, RPS23A [Ribosomal_proteins], RPS23B [Ribosomal_proteins], YCR025C, SRB7, EDC2, YAE1, MOH1, SWS2, GPI19, MTC7, MRPL51, DSS4
18.0-20.0 kDa	142	0	1	Actin_tubulin; Autophagy; Cell_cycle; ER_UPR; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	elevated	BUD25, YAR068W, YPL073C, IGO1, IRC23, YGL041W-A, YFL051C, GNA1, YHR180W, APS1, YPI1, FYV7, YLR281C, POP6, CAF20, RPL21A [Ribosomal_proteins], HED1, RPL21B [Ribosomal_proteins]
20.0-22.0 kDa	149	0	0	Actin_tubulin; Autophagy; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Mitochondrial_membrane; Proteasome; Ribosomal_proteins; Ubiquitin_system	elevated	YGR273C, FYV6, JJJ3, IRC25 [Heat_shock_chaperones; Proteasome], YGL242C, MRPL36, VPS68, UPS1, YOL024W, SCM4, YGL262W, YNL208W, MP01, FMP33, YKR045C, BNA1, MEH1, SDH4
22.0-24.0 kDa	154	0	0	Autophagy; Cell_cycle; ER_UPR; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Proteasome; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	elevated	FRQ1, COA1 [Mitochondrial_membrane], SRL1, FUN14, CIN4, SDH3, HAM1, ADD37, BET3, YOR387C, ERV2, YLR179C, VEL1, YDR248C, YMR295C, ATG31 [Autophagy], YPL257W, RPL16A [Ribosomal_proteins]
24.0-26.0 kDa	169	0	0	Autophagy; Cell_cycle; ER_UPR; Glycolytic_controls; Heat_shock_chaperones; Proteasome; Ribosomal_proteins; Ubiquitin_system	elevated	FMP32, FAR3, FAU1, ERP2, SLX9, SBA1 [Heat_shock_chaperones], ERV25 [ER_UPR], DOT5, RCN1, THO1, TRP1, MRX11, YEL076C-A, YLR464W, UBC1 [Ubiquitin_system], YBR197C, PSF1, RCR2
26.0-28.0 kDa	186	0	0	Cell_cycle; ER_UPR; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Proteasome; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	elevated	OCA6 [Protein_phosphatases], SNO4, CAT5, MGE1, TAE1, CYT2, MND1, YSA1, SHU2, YPT52, LSB1, SPR2, YKL033W-A, YOR283W, RIB5, TPT1, RTR1, VPS24
28.0-30.0 kDa	114	0	0	Actin_tubulin; Autophagy; Heat_shock_chaperones; Mitochondrial_membrane; Proteasome; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	elevated	SCL1 [Proteasome], ISY1, RPS0A [Ribosomal_proteins], YKL071W, IML3, USE1, RPL8B [Ribosomal_proteins], RPL8A [Ribosomal_proteins], LST7, MRPL15, RRP15, IES3, PSY3, SHE2, TAD2, RKI1, GRX3, PUP1 [Proteasome]
30.0-32.0 kDa	137	1	0	Actin_tubulin; Autophagy; Cell_cycle; Heat_shock_chaperones; Histones; Mitochondrial_membrane; Proteasome; Protein_phosphatases; Ubiquitin_system	high	ATO3, MRPL24, DC11, CYC3, MGM101, BMH1, SPO7, ERS1, PRO3, MAM33, DAL80, LUC7, YCL002C, ATG27 [Autophagy], PIR5, YCL068C, RSM26, SDH2
32.0-34.0 kDa	176	0	1	Actin_tubulin; Autophagy; Cell_cycle; ER_UPR; Glycolytic_controls; Heat_shock_chaperones; Mitochondrial_membrane; Ribosomal_proteins; Ubiquitin_system	high	YLR271W, TDA4, YFT2, PRM4, SNZ3, CRD1, VPS71, SNZ2, PEX18, PCL7 [Cell_cycle], LNP1 [ER_UPR], YMR087W, ASK1, PIN2, YNG2, INM2, ATP10 [Mitochondrial_membrane], PCL1 [Cell_cycle]
34.0-36.0 kDa	195	2	4	Actin_tubulin; Autophagy; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Mitochondrial_membrane; Proteasome; Protein_kinases; Protein_phosphatases; RAS_PKA_SNF1; TOR_AGC_signaling; Ubiquitin_system	high	ODC2, GCN3, TY3A-G, LST8 [TOR_AGC_signaling], COX11 [Mitochondrial_membrane], OAZ1, GID10, CYT1, CDC28 [Cell_cycle; Protein_kinases], YOR385W, CDC34 [Cell_cycle; Ubiquitin_system], SPC34, YNL194C, PHO4, SPE4, TY3A-I, AIM46, ECM11
36.0-38.0 kDa	149	1	0	Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Mitochondrial_membrane; Proteasome; Protein_kinases; Protein_phosphatases; RAS_PKA_SNF1	high	YMR262W, PHO92, YRB2, GPM2, YKL091C, OTU2, YKR075C, YPL039W, RRP36, RPC34, RFC4, IES2, LIA1, MRH1, YOR292C, RPA43, ELO1, MTD1
38.0-40.0 kDa	165	0	0	Actin_tubulin; Glycolytic_controls; Heat_shock_chaperones; Histones; Proteasome; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	elevated	SHE1, YML037C, LSP1, HUT1, ARI1, SOR2, TFB3, DFM1, SOR1, GRE2, YER156C, ARV1, SUA7, RFC3, YMR315W, ARA2, PHA2, GTT3
40.0-42.0 kDa	143	1	1	Actin_tubulin; Autophagy; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Histones; MAPK_signaling; Protein_kinases; Protein_phosphatases; Ubiquitin_system	high	ELO2, IBD2, RAV2, PDB1, RNR4, YML119W, LYS12, SAE2, YCR102C, DON1, RCL1, SCW4, DUG3, SRT1, TES1, URE2, MDG1, YBR053C
42.0-44.0 kDa	133	0	1	Actin_tubulin; Cell_cycle; Heat_shock_chaperones; Histones; MAPK_signaling; Mitochondrial_membrane; Proteasome; Protein_kinases; Protein_phosphatases; Ribosomal_proteins; Ubiquitin_system	high	PDE1, TOM40 [Mitochondrial_membrane], BMT6, MIG2, ETR1, CPR6, ARC1, RIM2, YOR338W, MTG1, SPG5, CAK1 [Protein_kinases], SAM2, SPC42, A15_BETA, PEX7, BCD1, TAF4
44.0-46.0 kDa	144	1	2	Actin_tubulin; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Histones; MAPK_signaling; Proteasome; Protein_kinases; Protein_phosphatases; RAS_PKA_SNF1	high	RRP43, TYS1, DOM34, ARP2 [Actin_tubulin], DCC1, SEN2, CNS1 [Heat_shock_chaperones], MVD1, PEX27, CLB6 [Cell_cycle], MRP4, OLA1, PTC4 [Protein_phosphatases], TAZ1, VMA5, TAM41, TPX2 [Protein_kinases; RAS_PKA_SNF1], SK18
46.0-48.0 kDa	143	0	1	Autophagy; Cell_cycle; ER_UPR; Glycolytic_controls; Histones; Proteasome; Protein_kinases; RAS_PKA_SNF1; TOR_AGC_signaling; Ubiquitin_system	high	KTR1, PSG1, AAT2, OPI1, TPX1 [Protein_kinases; RAS_PKA_SNF1], SPN1, CAR2, DMA1 [Ubiquitin_system], BDH2, NSE4, RNR2, YNL165W, RTC4, BAG7, INN1, SPE2, DDR48, ERV46
48.0-50.0 kDa	163	1	1	Actin_tubulin; Autophagy; Cell_cycle; Histones; MAPK_signaling; Mitochondrial_membrane; Proteasome; Protein_kinases; Ubiquitin_system	high	TY4A-J, SRO9, OSE3, ZNG1, DUS1, RMA1, COT1, PEA2, VPS4, PMI40, RAM1, SIM1, IDP1, SPR28, TY4A-H, MEI4, RPT5 [Proteasome], LIF1
50.0-52.0 kDa	143	1	0	Actin_tubulin; Autophagy; Cell_cycle; Heat_shock_chaperones; Histones; Proteasome; Protein_kinases; Protein_phosphatases	high	PZF1, UBX7, TEF1, ARP6 [Actin_tubulin], TRM11, RDS2, BNA3, RTT109 [Histones], DAL1, TEP1, RAX1, RNY1, RBA50, VBA3, CLP1, COR1, MOD5, CUS1
52.0-54.0 kDa	122	1	0	Actin_tubulin; Cell_cycle; Glycolytic_controls; Histones; Mitochondrial_membrane; Protein_kinases; Ubiquitin_system	high	SPS22, CLD1, YHR177W, CDC40 [Cell_cycle], YKL075C, YSP3, KTR6, VHS1 [Protein_kinases], HRB1, COX10 [Mitochondrial_membrane], SAD1, YLH47, SHM2, TMS1, YMR209C, RAD57, YPL245W, MCH2
54.0-56.0 kDa	122	0	2	Actin_tubulin; Autophagy; Cell_cycle; Heat_shock_chaperones; Histones; MAPK_signaling; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; Ubiquitin_system	high	LPD1, RNT1, OST1, GPA1, MST1, FAR10, TRM13, ITT1 [Ubiquitin_system], YMR265C, YMR027W, URN1, RTC6, NAR1, RSC6, APE4, TIM54 [Mitochondrial_membrane], UGA2
56.0-58.0 kDa	133	1	3	Autophagy; Cell_cycle; Glycolytic_controls; Heat_shock_chaperones; Histones; MAPK_signaling; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases; Ubiquitin_system	high	YHL012W, CYS4, ERG4, RUD3, DIT2, AMF1, RIO1 [Protein_kinases], TPS1, FDC1, RCM1, FMP42, ARO8, MMT1, MDM10, ARH1, CLB2 [Cell_cycle], STB3, EHD3
58.0-60.0 kDa	113	0	0	Autophagy; Cell_cycle; ER_UPR; Heat_shock_chaperones; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases	elevated	MGR3, CDC6 [Cell_cycle], ALG2, FCY21, TH16, GCR2, STP1, SDS23, HOM3, YJR015W [ER_UPR], GAL3, ANP1, FCY2, PLM2, DRN1, YPS6, PDI1 [ER_UPR], WSC3
60.0-62.0 kDa	102	0	0	Cell_cycle; Heat_shock_chaperones; Histones; Proteasome; Protein_kinases; Protein_phosphatases; TOR_AGC_signaling	elevated	YPS1, CHK1 [Protein_kinases], PTM1, YMR155W, BUD22, CDC3 [Cell_cycle], PAC11, TNA1, APE3, RPN4 [Proteasome], PNS1, YCK3, ERT1, RRN7, RPN3 [Proteasome], SRP1, MUD2, YBR138C
62.0-64.0 kDa	119	0	1	Cell_cycle; ER_UPR; Histones; Protein_kinases; Protein_phosphatases; Ubiquitin_system	elevated	CLN1 [Cell_cycle], HNM1, PCM1, GLE1, YCK2, TOS3 [Protein_kinases], DAK2, PUS1, MET7, DAK1, DED81, LCB1, PR12, YDL180W, UTP7, GIP2, GGA1, TDP1
64.0-66.0 kDa	86	0	0	Actin_tubulin; Cell_cycle; ER_UPR; Heat_shock_chaperones; Protein_kinases	routine	NSE5, ALP1, VPS36, YLR152C, TRM1, ERC1, MMP1, HSU1, MED1, TCM62 [Heat_shock_chaperones], MKC7, TPO1, RTC5, YAP1802, PAB1, GGA2, SAM3, PHO84
66.0-68.0 kDa	91	0	0	Actin_tubulin; Cell_cycle; Heat_shock_chaperones; Histones; Protein_kinases	routine	HIP1, PAP2, BFA1, HO, MET1, MSK1, APP1, DBF2 [Protein_kinases], PEX28, SPT8, DOT1 [Histones], RFT1, TDA1 [Protein_kinases], HXT5, STI1 [Heat_shock_chaperones], PHR1, BUD8, CK11
68.0-70.0 kDa	89	1	1	Heat_shock_chaperones; Protein_kinases; Protein_phosphatases	elevated	ECM3, PTR2, DBP9, RCK2 [Protein_kinases], TPO3, MMN4, LYP1, MAL12, DIP5, MAL32, PEX8, MPH2, RIM101, MAL31, YKR078W, CTM1, XKS1, RL11
70.0-72.0 kDa	81	0	0	Actin_tubulin; Autophagy; Heat_shock_chaperones; Mitochondrial_membrane; Protein_phosphatases; Ubiquitin_system	routine	ECM10 [Heat_shock_chaperones], TOM70 [Mitochondrial_membrane], PRM7, YHR219W, NHX1, SYO1, YSW1, SKO1, RLF2, LS86, YOR389W, SDH1, SAS10, SWP82, AEP3, SAG1, RFA1, GPI1
72.0-74.0 kDa	81	0	2	Actin_tubulin; Autophagy; Cell_cycle; Heat_shock_chaperones; MAPK_signaling; Protein_kinases; RAS_PKA_SNF1; Ubiquitin_system	elevated	UTP5, MDM32, CUE3, SNF1 [Protein_kinases; RAS_PKA_SNF1], ORC3, SPC72 [Actin_tubulin], SCC4, ELM1 [Protein_kinases], A15_ALPHA, FUI1, SPO1, RSM22, ALE1, MUB1, GRC3, RSC4, NEL1, MUS81
74.0-76.0 kDa	66	1	0	Cell_cycle; ER_UPR; Heat_shock_chaperones; Protein_kinases; Protein_phosphatases	elevated	SPH1, RPC82, ARE2, YNL115C, TPA1, NDC1, PIG1 [Protein_phosphatases], TRF5, MTO1, YJR098C, SGV1 [Protein_kinases], KAR9, CDC45 [Cell_cycle], YML082W, VPS52, TOP3, MEG2, MAE1
76.0-78.0 kDa	72	0	3	Heat_shock_chaperones; Protein_kinases; Protein_phosphatases; TOR_AGC_signaling; Ubiquitin_system	elevated	YML020W, CBP1, YPL109C, HOF1, YBL111C, MSS116, PTR3, CR21, MAG2 [Ubiquitin_system], TMN2, LAM5, BPL1, ALK2 [Protein_kinases], HEH2, RRI2, PSO2, PAN3, YPK1 [Protein_kinases; TOR_AGC_signaling]
78.0-80.0 kDa	56	0	0	Histones; Mitochondrial_membrane; Protein_kinases; Protein_phosphatases	routine	YDR444W, ROG1, GRS1, GAL10, LAM6, MLH2, GSH1, HCS1, TUP1, FMP40, YEL023C, DPB2, PPN1, KIP2, SRB4, PPZ2 [Protein_phosphatases], SSK1, IRC3

MW bin	N	Markers	Signal	Families	Priority	Examples / audit familles
80.0-82.0 kDa	62	1	1	Cell_cycle; Heat_shock_chaperones; Histones; MAPK_signaling; Protein_kinases; Ubiquitin_system	elevated	YGL114W, AVT4, MDV1, GFA1, HDA1 [Histones], FRE2, GSY2, RIM4, KRE6, YMR1, EXO1, UTP8, YOR365C, FRE5, YBR259W, APL1, NDC80, TRS85
82.0-84.0 kDa	45	0	0	Autophagy; Heat_shock_chaperones; Protein_kinases; Ubiquitin_system	routine	MLH3, FRE4, RAD34, BCH1, NUP82, GYL1, ESC8, TFG1, RGD2, KEX1, YKL222C, OSW2, DSF2, SEC39, CLF1, IML2, KKQ8 [Protein_kinases], VRP1
84.0-86.0 kDa	59	0	1	Cell_cycle; Histones; Protein_kinases; Protein_phosphatases	routine	KAR3, RRP6, POX1, SEC18, YNL144C, CTF3, YJR030C, MAK10, SUV3, YOL036W, SLX4, CTF18, SET2 [Histones], YJL132W, GID7, THS1, AFG3, MEF1
86.0-88.0 kDa	55	0	0	Actin_tubulin; Cell_cycle; Protein_kinases; Ubiquitin_system	routine	NTO1, CHS6, ADE57, ALK1 [Protein_kinases], CDC4 [Cell_cycle], CUL3, BRN1, SKN1, UBP9 [Ubiquitin_system], PKH1 [Protein_kinases], PMT3, PBY1 [Actin_tubulin], TOF2, MCM5, LCD1, AVL9, YDR239C, NSP1
88.0-90.0 kDa	39	0	0	Protein_phosphatases; TOR_AGC_signaling; Ubiquitin_system	routine	PMT6, BCH2, DSL1, PRT1, OCT1, ACF2, PTA1, PAT1, MNN1, UBP10 [Ubiquitin_system], GAC1 [Protein_phosphatases], ABZ1, UBP14 [Ubiquitin_system], URC2, TRE1, BUD6, STB6, TCO89 [TOR_AGC_signaling]
90.0-92.0 kDa	41	0	1	Actin_tubulin; Cell_cycle; Heat_shock_chaperones; Protein_kinases; Protein_phosphatases; RAS_PKA_SNF1; TOR_AGC_signaling; Ubiquitin_system	elevated	KEX2, KIN4 [Protein_kinases], RGD3, RPH1, NAB3, SWI6, VPS72, RFX1, TSR1, FLC3, AXL2, FYV8, IOC3, SCY1 [Protein_kinases], UTP13, PRK1 [Actin_tubulin], SIP1 [Protein_kinases; RAS_PKA_SNF1], TOG1
92.0-94.0 kDa	39	0	0	Cell_cycle; Heat_shock_chaperones; Protein_kinases; Ubiquitin_system	routine	VPS70, SVL3, YGR067C, UBP5 [Ubiquitin_system], ROD1, RAP1, YRR1, SWE1 [Protein_kinases], COG3, PMT1, YGR130C, VPS16, UBP1 [Ubiquitin_system], MDM20, PAM1, YBR284W, SWI3, RIX7
94.0-96.0 kDa	39	1	0	Cell_cycle; Heat_shock_chaperones; Protein_kinases; Ubiquitin_system	elevated	NUD1, YMR102C, NOP14, SYH1, FKH2, NOT3, MKT1, PET111, FAR1 [Cell_cycle; Protein_kinases], YLL032C, CCR4, HPF1, SAP4, KAP95, LAS21, ARG5.6, RFC1, MCM7
96.0-98.0 kDa	40	0	0	Cell_cycle; ER_UPR; Heat_shock_chaperones; Histones; Mitochondrial_membrane	routine	SIP4, DGR2, AII [Mitochondrial_membrane], YHR131C, SYP1, NIC96, CFT2, PRP5, XRS2, SBE22, SPB1, CWH41, VTC3 [Heat_shock_chaperones], USA1, ALR2, YME2, SNF3, EPL1
98.0-100.0 kDa	39	1	0	Heat_shock_chaperones; Mitochondrial_membrane; Protein_phosphatases	elevated	A12 [Mitochondrial_membrane], PSY2 [Protein_phosphatases], PUF3, DUG2, ENV11, SPC98, SND1, HIR2, GPB2, COG4, JIP4, FOX2, VTC5 [Heat_shock_chaperones], MCM2, CHL1, FIR1, ZAP1, NAT1
100.0-102.0 kDa	43	0	0	Actin_tubulin; Autophagy; Protein_kinases; Ubiquitin_system	routine	STO1, PXA1, ZIP1, LEU3, ARP8 [Actin_tubulin], SYF1, ARG81, NIP100, SAP1, UPC2, SEC10, POP1, PKH3 [Protein_kinases], FPK1, PLC1, BNI4, CVK3, YPL150W [Protein_kinases]
102.0-104.0 kDa	39	0	1	Heat_shock_chaperones; MAPK_signaling; Protein_kinases; Ubiquitin_system	routine	HSP104 [Heat_shock_chaperones], RRN6, SPO75, PMA2, ADE3, MSH5, AFI1, RSC2, EMW1, STE20 [MAPK_signaling; Protein_kinases], VIR1, VNX1, VID22, PHO87, ALY1, SN5, YPR089W, EST2
104.0-106.0 kDa	28	0	0	Cell_cycle; Proteasome; Protein_phosphatases; Ubiquitin_system	routine	SYG1, PRP6, RPN2 [Proteasome], YJL070C, AGC1, ORC1, CTF4, PMR1, COS111, PFK2, SOV1, BUL2 [Ubiquitin_system], UTP21, SPC105, SEC21, CDC13 [Cell_cycle], MCM4, SEC15
106.0-108.0 kDa	18	0	0		routine	YEL043W, MIS1, DIF2, NNF2, RSC1, APL5, TIF4631, STE13, CYC8, PEP3, MCM3, WAR1, RSN1, TMA108, APE2, CWH43, RDH54, PFK1
108.0-110.0 kDa	31	0	0	Proteasome; Ubiquitin_system	routine	VID30, ARO80, SXM1, MNN14, INP51, FAP1, DNL4, EGT2, DCP2, BIR1, ASG1, MSH2, SLA2, SEC26, VPS35, DAL81, BUL1 [Ubiquitin_system], MDR1
110.0-112.0 kDa	17	0	0	Cell_cycle	routine	HSH155, ALA1, CDC15 [Cell_cycle], ROT2, RPG1, GPR1, MTR10, DSS1, NMA111, PFF1, SIN4, SIR3, PUT3, SRO77, SPC110, CBF2, FLO5
112.0-114.0 kDa	25	0	0	Mitochondrial_membrane	routine	SEC5, CYM1, REV1, FUN12, SPO22, EAF1, PHM7, PDR3, REG1, PET309 [Mitochondrial_membrane], RGA1, DBP10, MCM6, SFI1, YMR317W, RTP1, RGA2, CIN8
114.0-116.0 kDa	26	0	0	Autophagy; Ubiquitin_system	routine	SNU114, MPH1, SED4, SKG3, UBA1 [Ubiquitin_system], KGD1, GCV2, RTT10, SRO7, PIP2, MET10, SAP155, APL3, YPR204W, ATG9 [Autophagy], PEX6, HMG1, GPI13
116.0-118.0 kDa	24	0	0	Protein_kinases; Ubiquitin_system	routine	NUP116, KCC4 [Protein_kinases], MSH3, CIN1, LRG1, MAK21, ULP2 [Ubiquitin_system], VPS3, VSB1, KIC1 [Protein_kinases], KSP1 [Protein_kinases], SAP190, ALY2, ADP1, PEX1, PEP5 [Ubiquitin_system], STE23, SE44
118.0-120.0 kDa	18	0	0		routine	SUM1, CAF120, DAN4, YOL098C, SPT16, VMA1, TPS3, TT1, OAF1, RQC2, KRE33, ZRG8, PUF2, MIL1, KCS1, KAP120, PIK1, NMD5
120.0-122.0 kDa	18	0	0	Protein_kinases	routine	KIN1 [Protein_kinases], TFC4, ENA5, GAL11, ENA2, ENA1, JSN1, RGC1, NUP120, TAF1, PSE1, DSE4, SPT23, SAP185, RIC1, PRP16, PKH2 [Protein_kinases], PDR1
122.0-124.0 kDa	21	0	0	Histones; Protein_kinases; Ubiquitin_system	routine	MTR4, FLO10, SEC8, IFH1, KAP123, VAM6, ILS1, RTT107, TSL1, LEC1, UBP7 [Ubiquitin_system], RGR1, KAP122, HRQ1, HOS4, ECM21, SWI4, SET1 [Histones]
124.0-126.0 kDa	18	0	0	Cell_cycle; Protein_kinases	routine	CRM1, CDC60 [Cell_cycle], MRC1, GDH2, PSK2 [Protein_kinases], RIA1, AMS1, RAD26, INP53, POL3, CTR9, CSR2, BEM3, YNR065C, DUF1, SEG2, VAS1, KIP1
126.0-128.0 kDa	18	0	0	ER_UPR; Protein_kinases; RAS_PKA_SNF1	routine	SMC5, NAB6, RAD1, YMR196W, BUD2, NMD2, LOS1, ASK10, SAK1 [RAS_PKA_SNF1], TBS1, ECM32, IRE1 [ER_UPR; Protein_kinases], PAN2, MGA2, PIM1, TY4B-P, YJR039W, SEF1
128.0-130.0 kDa	20	0	0	Protein_kinases	routine	SMC6, TRS130, YPL216W, VAC7, RGT1, BBC1, KIN2 [Protein_kinases], FUN30, NET1, ESL1, CAF130, NUT1, RET1, BRE4, VIP1, GIN4 [Protein_kinases], CHS1, MSB1
130.0-132.0 kDa	16	0	1	Protein_kinases; TOR_AGC_signaling	routine	PRP22, PSD2, PYC1, YPR022C, PYC2, NEO1, ISW2, PMC1, MTC5, KEL1, ISW1, AVO1 [TOR_AGC_signaling], ROM1, PKC1 [Protein_kinases], CHS3, PCA1
132.0-134.0 kDa	12	0	0	Ubiquitin_system	routine	TFC3, TCB2, NCR1, GRR1 [Ubiquitin_system], YCS4, IRR1, MSB2, NUP133, INP52, TCB1, SMC2, RAX2
134.0-136.0 kDa	15	0	0	Actin_tubulin; Autophagy; ER_UPR; Protein_kinases; Ubiquitin_system	routine	RAD5 [Ubiquitin_system], PHO81, SRS2, NEW1, SLN1 [Protein_kinases], SCP160, ATG11 [Autophagy], HFM1, SWH1, URB2, SPF1 [ER_UPR], BMS1, COP1, RPA135, SLA1 [Actin_tubulin]
136.0-138.0 kDa	12	0	0	Actin_tubulin; Autophagy	routine	ATG26 [Autophagy], YEL025C, FLO11, TY18-A, MYO5, ATH1, YGL140C, ESL2, RRP12, SYT1, AIP5 [Actin_tubulin], NTF1
138.0-140.0 kDa	10	0	0		routine	GDE1, FLO9, AXL1, SNT1, SEC31, RPB2, RPM2, MNN4, HER1, SSD1
140.0-142.0 kDa	10	0	0		routine	MSH6, OXP1, UTP22, GIP3, TRK1, TOF1, SMC1, SMC3, TDA9, NST1
142.0-144.0 kDa	8	0	0	Ubiquitin_system	routine	MSN5, MYO3, SIP3, YEL077C, UBP12 [Ubiquitin_system], MIP1, UBP15 [Ubiquitin_system], LAM1
144.0-146.0 kDa	10	0	0		routine	STE6, ECM16, VPS8, ECM30, YOL075C, CLU1, SPO71, ITC1, NUP145, OSH2
146.0-148.0 kDa	6	0	0	Ubiquitin_system	routine	SKI2, UBP2 [Ubiquitin_system], PDS5, YOR296W, TRS120, SWI1
148.0-150.0 kDa	7	0	0		routine	RAD9, ADE6, TUS1, RTC1, SAC3, LAM4, PDR18
150.0-152.0 kDa	5	0	0	Ubiquitin_system	routine	EDE1, ADR1, YLR278C, SSM4 [Ubiquitin_system], POM152
152.0-154.0 kDa	11	0	0	Protein_kinases	routine	SIR4, PSK1 [Protein_kinases], RAD50, ROM2, SPT7, SSK22 [Protein_kinases], IKI3, RPO41, CFT1, DRS2, RSE1
154.0-156.0 kDa	8	0	0		routine	SEC3, RAV1, YLL067C, RSF2, MHP1, LYS2, YLL066C, YML133C
156.0-158.0 kDa	7	0	0	Heat_shock_chaperones; Histones	routine	YRF1-4, KRE5, NUP157, STH1, BNR1, YTA7 [Heat_shock_chaperones; Histones], AUS1
158.0-160.0 kDa	2	0	0		routine	NUP159, GEA1
160.0-162.0 kDa	10	0	0	Actin_tubulin; Ubiquitin_system	routine	SSN2, PAN1 [Actin_tubulin], CAT8, PDR11, YSP2, FLO1, MET5, MMS1 [Ubiquitin_system], TAF2, SPP41
162.0-164.0 kDa	10	0	0	Ubiquitin_system	routine	SMC4, RPO31, ECM5, YLR419W, SPA2, LTE1, SNT2 [Ubiquitin_system], TRM732, SKI3, SGI1
164.0-166.0 kDa	6	0	1	MAPK_signaling; Protein_kinases; TOR_AGC_signaling	routine	BCK1 [MAPK_signaling; Protein_kinases], TOP2, TSC11 [TOR_AGC_signaling], BUD4, TRM3, GEA2
166.0-168.0 kDa	10	0	0	Protein_kinases; Ubiquitin_system	routine	FIG2, HAP1, VPS15 [Protein_kinases], YOR1, YPK9, POL1, SRB8, MDS3, MMS22 [Ubiquitin_system], UFD4 [Ubiquitin_system]
168.0-170.0 kDa	6	0	0	Protein_kinases	routine	CHD1, SPT6, SNQ2, MYO4, NUP170, HSL1 [Protein_kinases]
170.0-172.0 kDa	7	0	0		routine	PDR5, PDR12, TCB3, SCC2, YCF1, INO80, DNA2
172.0-174.0 kDa	4	0	0		routine	PDR15, TY3B-I, IQG1, REV3
174.0-176.0 kDa	8	0	0		routine	STU1, VTH2, VTH1, SWR1, ARO1, SIN3, GDB1, XRN1
176.0-178.0 kDa	5	0	0	TOR_AGC_signaling	routine	PDR10, BPT1, KOG1 [TOR_AGC_signaling], PEP1, DNF1
178.0-180.0 kDa	3	0	0	Autophagy; Cell_cycle	routine	TY3B-G, ATG2 [Autophagy], CDC25 [Cell_cycle]
180.0-182.0 kDa	7	0	0	MAPK_signaling; Ubiquitin_system	routine	RKR1 [Ubiquitin_system], IRC20 [Ubiquitin_system], SSK2 [MAPK_signaling], MYO2, VMR1, TY1B-DR4, IML1
182.0-184.0 kDa	2	0	0		routine	DNF2, THO2
184.0-186.0 kDa	2	0	0	Ubiquitin_system	routine	ULS1 [Ubiquitin_system], BUD3
186.0-188.0 kDa	7	0	0		routine	RPA190, MON2, CRM2, NTE1, ESC1, CHC1, ESP1
188.0-190.0 kDa	4	0	0		routine	DNF3, NUP188, HKR1, YBT1
190.0-192.0 kDa	5	0	1	Histones	routine	GCN2, YRF1-2, NUP192, RPO21, HIR3 [Histones]
192.0-194.0 kDa	1	0	0		routine	RRP5
194.0-196.0 kDa	5	0	0		routine	SNF2, DOP1, MLP2, SPO14, PMD1
196.0-198.0 kDa	4	0	1	Protein_kinases	routine	APC1, RIM15 [Protein_kinases], YIL177C, YJL225C
198.0-200.0 kDa	28	0	0		routine	TY1B-NL2, TY1B-PR3, TY1B-LR3, TY1B-LR1, TY1B-GR2, TY1B-ML1, TY1B-DR5, TY1B-OR, TY1B-LR2, TY1B-MR2, TY1B-GR3, TY1B-ER1, TY1B-PR1, TY1B-ER2, TY1B-ML2, TY1B-PL, TY1B-DR1, TY1B-OL
200.0-202.0 kDa	5	0	0		routine	UTP10, DURL2, TY2B-DR3, TY2B-DR1, TY2B-OR1
202.0-204.0 kDa	12	0	0		routine	TY2B-F, TY2B-GR2, TY2B-DR2, TY2B-C, TY2B-LR1, TY2B-OR2, TY2B-B, TY1B-H, URB1, YRF1-1, YRF1-5, YRF1-8
206.0-208.0 kDa	5	0	0		routine	USO1, FAS2, FKS3, TY4B-J, TY4B-H
208.0-210.0 kDa	1	0	0		routine	MOT1
210.0-212.0 kDa	4	0	0	Proteasome	routine	ECM29 [Proteasome], YRF1-6, YRF1-3, YRF1-7

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
214.0-216.0 kDa	2	0	0		routine	STT4, FKS1
216.0-218.0 kDa	3	0	0	Ubiquitin_system	routine	UBR2 [Ubiquitin_system], GSC2, RIF1
218.0-220.0 kDa	2	0	0		routine	MLP1, BNI1
220.0-222.0 kDa	1	0	0		routine	DCK1
222.0-224.0 kDa	1	0	0		routine	MYO1
224.0-226.0 kDa	2	0	0	Ubiquitin_system	routine	UBR1 [Ubiquitin_system], SLH1
226.0-228.0 kDa	2	0	0		routine	SEC7, CYR1
228.0-230.0 kDa	2	0	0		routine	FAS1, LAA1
234.0-236.0 kDa	1	0	0		routine	CCH1
238.0-240.0 kDa	1	0	0		routine	GLT1
240.0-242.0 kDa	2	0	0	Cell_cycle	routine	CDC39 [Cell_cycle], SEC16
244.0-246.0 kDa	3	0	0	Proteasome	routine	URA2, BEM2, BLM10 [Proteasome]
246.0-248.0 kDa	1	0	0		routine	BRR2
250.0-252.0 kDa	2	0	0		routine	ACC1, BPH1
252.0-254.0 kDa	1	0	0		routine	SEN1
254.0-256.0 kDa	1	0	0		routine	POL2
256.0-258.0 kDa	1	0	0		routine	FAB1
258.0-260.0 kDa	1	0	0		routine	HFA1
268.0-270.0 kDa	1	0	0		routine	TAO3
272.0-274.0 kDa	1	0	0	Protein_kinases	routine	MEC1 [Protein_kinases]
278.0-280.0 kDa	1	0	0		routine	PRP8
280.0-282.0 kDa	2	0	2	Protein_kinases; TOR_AGC_signaling	routine	TOR1 [Protein_kinases; TOR_AGC_signaling], TOR2 [Protein_kinases; TOR_AGC_signaling]
284.0-286.0 kDa	1	0	0		routine	HOB2
286.0-288.0 kDa	1	0	0		routine	UTP20
296.0-298.0 kDa	1	0	0		routine	GCN1
302.0-304.0 kDa	1	0	0		routine	FMP27
312.0-314.0 kDa	1	0	0		routine	NUM1
320.0-322.0 kDa	1	0	0	Protein_kinases	routine	TEL1 [Protein_kinases]
338.0-340.0 kDa	1	0	0		routine	CSF1
350.0-352.0 kDa	2	0	0		routine	IRA1, IRA2
356.0-358.0 kDa	1	0	0		routine	VPS13
374.0-376.0 kDa	1	0	0	Mitochondrial_membrane; Ubiquitin_system	routine	TOM1 [Mitochondrial_membrane; Ubiquitin_system]
432.0-434.0 kDa	1	0	0		routine	TRA1
470.0-472.0 kDa	1	0	0		routine	DYN1
558.0-560.0 kDa	1	0	0		routine	MDN1

Proteins sorted from lowest to highest molecular weight

The PDF shows the first 120 proteins. The complete table is in proteome_wb_integrity_atlas.csv.

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
1.98	YJR151W-A	Q3E775	Uncharacterized protein YJR151W-A	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.71	YNL040C-A	P9WEJ1	Uncharacterized protein YNL040C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.98	YOR302W	P08521	Arginine attenuator peptide	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.07	YPR160W-A	P0C5S1	Uncharacterized protein YPR160W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.07	YHR073C-B	P0C5N4	Uncharacterized protein YHR073C-B	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.08	YJL077W-A	Q8TGN3	Uncharacterized protein YJL077W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.09	YMR182W-A	Q8TGS7	Uncharacterized protein YMR182W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.10	YMR242W-A	Q8TGS6	Uncharacterized protein YMR242W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.14	YJL136W-A	Q3E801	Uncharacterized protein YJL136W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.17	YGL007C-A	Q8TGU0	Uncharacterized protein YGL007C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.20	YHL015W-A	Q3E726	Uncharacterized protein YHL015W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.32	YAR035C-A	Q8TGV0	Uncharacterized protein YAR035C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.34	RPL41A	P0CX86	Small ribosomal subunit protein eS32A	2.0-4.0 kDa	Ribosomal_proteins	configured audit family match: Ribosomal_proteins
3.34	RPL41B	P0CX87	Small ribosomal subunit protein eS32B	2.0-4.0 kDa	Ribosomal_proteins	configured audit family match: Ribosomal_proteins
3.34	YKL145W-A	P0C5P3	Uncharacterized protein YKL145W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.37	YNR075C-A	Q8TGJ2	Putative UPF0377 protein YNR075C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.38	YFR012W-A	Q3E838	Uncharacterized protein YFR012W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.39	YGR174W-A	Q8TGT8	Uncharacterized protein YGR174W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.40	YBL039C-A	P0C5K9	Uncharacterized protein YBL039C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.41	YKL100W-A	P0C5P2	Uncharacterized protein YKL100W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.42	YER090C-A	P0C5M8	Uncharacterized protein YER090C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.45	YAL037C-A	Q3E741	Putative uncharacterized protein YAL037C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.47	RDT1	A0A0B7P221	Uncharacterized protein RDT1	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.51	YHR022C-A	Q8TGT6	Uncharacterized protein YHR022C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.53	YDR524W-C	P0C1Z1	Uncharacterized protein YDR524W-C	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.58	YPR159C-A	Q8TGQ7	Uncharacterized protein YPR159C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.62	YOL038C-A	Q3E729	Uncharacterized protein YOL038C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.69	YDR371C-A	P0C5M1	Uncharacterized protein YDR371C-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.84	YPL152W-A	Q8TGR9	Uncharacterized protein YPL152W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.85	YBL071C-B	Q8TGU8	Uncharacterized protein YBL071C-B	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.92	YBR196C-B	Q3E778	Uncharacterized protein YBR196C-B	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.93	MFA1	P34165	Mating hormone A-factor 1	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.94	YLR163W-A	P0C5P6	Uncharacterized protein YLR163W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.95	YER138W-A	P0C270	Putative uncharacterized protein YER138W-A	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.98	OST4	Q99380	Dolichyl-diphosphooligosaccharide-protein glycosyltransferase subunit OST4	2.0-4.0 kDa		standard full-blot and antibody-validation checks
4.01	YJL077W-B	Q8TGT3	Uncharacterized protein YJL077W-B	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.05	YMR315W-A	Q8TGS4	Uncharacterized protein YMR315W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.09	YBR221W-A	Q3E781	Uncharacterized protein YBR221W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.11	YMR272W-B	Q8TGS5	Uncharacterized protein YMR272W-B	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.21	YGL006W-A	Q3E802	Uncharacterized protein YGL006W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.23	MFA2	P34166	Mating hormone A-factor 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.41	YML007C-A	Q3E7A6	Uncharacterized protein YML007C-A, mitochondrial	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.45	YDR118W-A	P0C5L7	Uncharacterized protein YDR118W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.46	YER053C-A	Q3E7B0	Uncharacterized protein YER053C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.50	PMP1	P32903	Plasma membrane ATPase proteolipid 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.51	YBR296C-A	Q8TGU5	Uncharacterized protein YBR296C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.52	YLR264C-A	Q3E732	Putative uncharacterized protein YLR264C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.54	YGR204C-A	Q8TGT7	Uncharacterized protein YGR204C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.55	YJL052C-A	Q3E837	Uncharacterized protein YJL052C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.62	PLS1	P0C271	Peroxisomal LYS1 stabilizing protein 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.64	YBL100W-C	P0C269	Uncharacterized protein YBL100W-C	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.68	YKL106C-A	Q3E7A0	Uncharacterized protein YKL106C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.68	YLR364C-A	P0C5Q1	Uncharacterized protein YLR364C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.71	YLR154C-H	P0CY02	Uncharacterized protein YLR154C-H	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.71	YLR156C-A	P0CY03	Uncharacterized protein YLR156C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.71	YLR157C-C	P0CY04	Uncharacterized protein YLR157C-C	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.71	YLR159C-A	P0CY05	Uncharacterized protein YLR159C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.84	YDR003W-A	Q3E727	Uncharacterized protein YDR003W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.85	PMP2	P40975	Plasma membrane ATPase proteolipid 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.05	YJL047C-A	Q3E737	Uncharacterized protein YJL047C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.09	YDL159W-A	Q3E774	Uncharacterized protein YDL159W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.10	YHL048C-A	Q3E758	Uncharacterized protein YHL048C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.15	YMR247W-A	Q3E782	Uncharacterized protein YMR247W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.20	YOL155W-A	Q3E7Y8	Uncharacterized protein YOL155W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.22	YIR018C-A	Q3E723	Uncharacterized protein YIR018C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.28	YMR158C-A	Q3E843	Uncharacterized protein YMR158C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.31	YGL188C-A	Q8TGU1	Uncharacterized protein YGL188C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.32	YNL067W-B	Q3E767	Uncharacterized protein YNL067W-B	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.33	YDR320W-B	P0C5M0	Uncharacterized protein YDR320W-B	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.43	YOR161C-C	Q8TGS0	Uncharacterized protein YOR161C-C	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.57	HOR7	Q05827	Protein HOR7	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.59	YDR169C-A	Q3E789	Uncharacterized protein YDR169C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.61	DGR1	Q3E808	2-deoxy-glucose resistant protein 1, mitochondrial	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.63	YOL083C-A	P0C5R4	Uncharacterized protein YOL083C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.67	PAU7	P39545	Seripauperin-7	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.70	YMR013C-A	P0C5Q3	Uncharacterized protein YMR013C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.71	YOL019W-A	Q8TGS2	Uncharacterized protein YOL019W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.71	YHR175W-A	Q3E815	Uncharacterized membrane protein YHR175W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
5.80	YDL204W-A	P9WEJ5	Uncharacterized protein YDL204W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.81	YLR406C-A	Q8TGT1	Uncharacterized protein YLR406C-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.82	ATP8	P00856	ATP synthase protein 8	4.0-6.0 kDa	Mitochondrial_membrane	configured audit family match: Mitochondrial_membrane
5.85	Q0092	Q35811	Putative uncharacterized protein Q0092, mitochondrial	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.87	YHR052C-B	P9WEN8	Uncharacterized protein YHR052C-B	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.87	YHR054C-B	P9WEN9	Uncharacterized protein YHR054C-B	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.91	Q0297	Q9ZZV8	Putative uncharacterized protein Q0297, mitochondrial	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.93	YIL105W-A	Q8TGN5	Uncharacterized protein YIL105W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.94	Q0143	Q9ZWV3	Putative uncharacterized protein Q0143, mitochondrial	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.95	YOL166W-A	Q8TGJ1	UPF0320 protein YOL166W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.95	DDR2	P89113	Protein DDR2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.97	CPP3	Q6Q5X2	Lipid-anchored plasma membrane protein CPP3	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.98	TOM5	P80967	Mitochondrial import receptor subunit TOM5	4.0-6.0 kDa	Mitochondrial_membrane	configured audit family match: Mitochondrial_membrane
5.99	YHR086W-A	Q3E746	Uncharacterized protein YHR086W-A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.99	YKL096C-B	Q8TGT2	Uncharacterized protein YKL096C-B	4.0-6.0 kDa		standard full-blot and antibody-validation checks
6.02	MCO6	Q3E828	ERMES regulator 1	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.05	YLR154C-G	Q3E813	Uncharacterized protein YLR154C-G	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.05	YGR146C-A	Q8TGT9	Uncharacterized protein YGR146C-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.10	YML054C-A	Q3E809	Uncharacterized protein YML054C-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.12	YOR293C-A	Q3E7Y7	Uncharacterized protein YOR293C-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.13	YDR194W-A	Q3E818	Uncharacterized protein YDR194W-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.13	YOR072W-B	Q3E832	Uncharacterized protein YOR072W-B	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.14	PMP3	P87284	Plasma membrane proteolipid 3	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.14	YBR072C-A	Q3E794	Uncharacterized protein YBR072C-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.14	YLR157W-E	P0CF00	Putative uncharacterized protein YLR157W-E	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.21	YOR376W-A	Q3E805	Uncharacterized protein YOR376W-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.23	YBR121C-A	P0C5L4	Uncharacterized protein YBR121C-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.25	YML100W-A	Q8TGT0	Uncharacterized protein YML100W-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.26	YBR196C-A	Q3E820	Uncharacterized protein YBR196C-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.26	YER078W-A	Q3E7A2	Uncharacterized protein YER078W-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.27	Q0017	Q9ZZX8	Putative uncharacterized protein Q0017, mitochondrial	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.30	YCR095W-A	Q3E787	Putative uncharacterized protein YCR095W-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.31	YIL046W-A	Q3E7Z4	Uncharacterized protein YIL046W-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.31	YOR394C-A	Q8TGJ0	Uncharacterized protein YOR394C-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.32	YDR524C-B	Q3E6R4	Uncharacterized protein YDR524C-B	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.34	NCE101	Q02820	Non-classical export protein 1	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.34	RPL39	P04650	Large ribosomal subunit protein eL39	6.0-8.0 kDa	Ribosomal_proteins	configured audit family match: Ribosomal_proteins
6.41	TOM6	P33448	Mitochondrial import receptor subunit TOM6	6.0-8.0 kDa	Mitochondrial_membrane	configured audit family match: Mitochondrial_membrane
6.41	YLL066W-B	Q8TGJ7	Uncharacterized protein YLL066W-B	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.46	YBR200W-A	Q3E755	Uncharacterized protein YBR200W-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.47	YER175W-A	Q8TGU4	Uncharacterized protein YER175W-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.51	YDR034C-A	P0C289	Putative uncharacterized protein YDR034C-A	6.0-8.0 kDa		standard full-blot and antibody-validation checks

Reported western-blot malpractice cases mapped to MW bins

These are reported public cases supplied in the malpractice input TSV. They are evidence examples for integrity-audit results sections, not substitution recommendations.

Case	Claim/protein	MW bin	Reported issue	Results sentence
YEAST_Lange_PNAS2000_Leu1	Leu1p western blot	84.0-86.0 kDa	PNAS correction states that the thermoprinter-paper Western-blot printout was cut into pieces and reassembled because gel loading did not match the desired presentation order; the published panel also contained an unintentional mirrored duplication of the 64 h bands.	A PNAS correction documents cut-and-reassembled Leu1p Western-blot lanes and a mirrored duplication in a direct <i>S. cerevisiae</i> experiment, mapping this audit precedent to the approximately 86 kDa neighborhood.
YEAST_Schuck_JCB2009_Rtn1Cherry	Rtn1-Cherry western blot detected through the Cherry tag	58.0-60.0 kDa	JCB correction states that Fig. 7A was assembled incorrectly, resulting in inadvertent duplication of lanes showing Rtn1 in DTT-treated cells.	A JCB correction documents duplicated Rtn1-Cherry lanes in a direct <i>S. cerevisiae</i> Western blot, providing an audit precedent around the roughly 60 kDa fusion-protein neighborhood.
YEAST_Schuck_JCB2009_Pgk1	Pgk1 loading-control western blot	44.0-46.0 kDa	JCB correction states that Fig. 7A was assembled incorrectly, resulting in mislabeling of the lanes showing the Pgk1 loading control.	The same JCB correction documents mislabeled Pgk1 loading-control lanes in a direct <i>S. cerevisiae</i> Western blot, mapping the corrected control to the approximately 45 kDa neighborhood.

Integrity-audit checklist

- Ask for the full, uncropped blot with ladder and lane labels. Cropped panels are not enough.
- Confirm expected molecular weight and apparent molecular weight. Do not treat molecular-weight agreement as proof of identity.
- For configured marker proteins, request source-imager metadata and original antibody-validation records.
- Check whether a marker band is near common markers such as GAPDH, actin, tubulin, HSPs, lamins, VDAC, PCNA, or other configured controls.
- For signaling proteins, verify phospho/total controls, controls for activation state, and an independent identity control when central to the claim.
- In crowded or high-audit bins, prioritize raw-data provenance, full blot context, and orthogonal validation.
- Interpret mapped malpractice cases only as documented audit precedents. They do not imply that proteins in the same bin are interchangeable.

Generated files

- proteome_wb_integrity_atlas.csv - complete per-protein atlas sorted by theoretical MW.
- mw_bin_summary.csv - complete molecular-weight-bin audit summary.
- reported_malpractice_bins.csv - generated when a malpractice TSV is supplied.
- plots/ - four standalone PNG figures embedded in this report.
- western_blot_integrity_atlas_full_report_with_plots.pdf - this consolidated report.
- western_blot_integrity_atlas_report.pdf - byte-identical compatibility copy of the consolidated report.

Input configuration

Input	File
Proteome FASTA	UP000002311_559292.fasta
Marker file	yeast_marker_genes.tsv
Signaling-prefix file	yeast_signaling_prefixes.tsv
Family-pattern file	yeast_family_patterns.tsv
Species/report label	Saccharomyces cerevisiae (S288c)